



Blacktown Boys' High School

2026

HSC Trial Examination

Mathematics Extension 1

**General
Instructions**

- Reading time – 10 minutes
- Working time – 2 hours
- Write using black pen
- Calculators approved by NESA may be used
- A reference sheet is provided for this paper
- All diagrams are not drawn to scale
- In Questions 11 – 14, show relevant mathematical reasoning and/or calculations

Total marks: **Section I – 10 marks** (pages 3 – 6)
70

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section

Section II – 60 marks (pages 7 – 12)

- Attempt Questions 11 – 14
- Allow about 1 hour and 45 minutes for this section

Assessor: X. Chirgwin

Student Name: _____

Teacher Name: _____

Students are advised that this is a trial examination only and cannot in any way guarantee the content or format of the 2026 Higher School Certificate Examination.

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Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the multiple choice answer sheet for Questions 1–10.

Q1. What is the range of the function $f(x) = \tan^{-1}(2x) - \pi$?

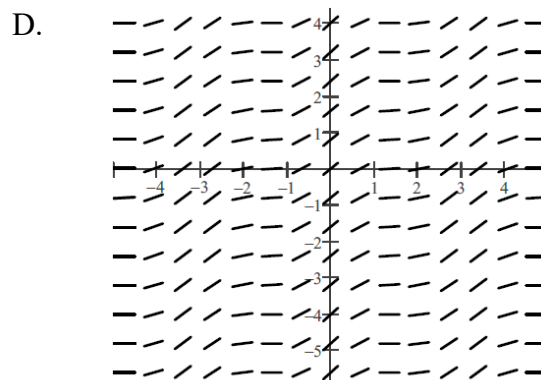
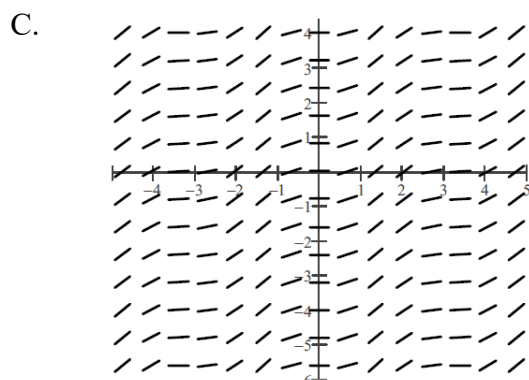
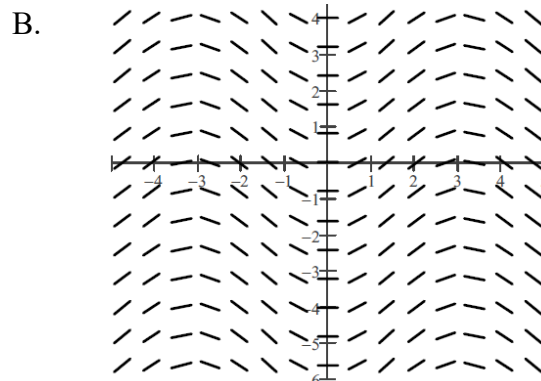
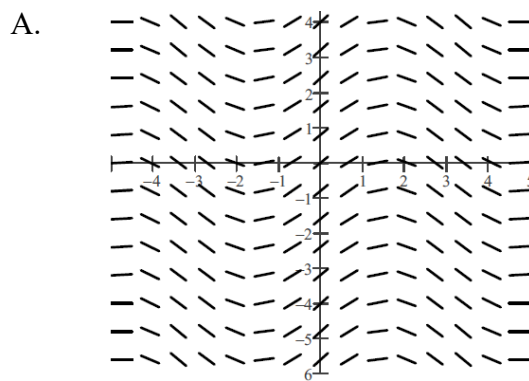
A. $-\pi < y < 0$

B. $-\frac{\pi}{2} < y < \frac{\pi}{2}$

C. $-\frac{3\pi}{2} < y < -\frac{\pi}{2}$

D. $-2\pi < y < -\pi$

Q2. Which of the following best represents the slope field for $\frac{dy}{dx} = \cos x$?



Q3. A Bernoulli random variable X has probability distribution

$$P(x) = \frac{3x + 1}{5} \quad \text{for } x = 0, 1.$$

What are the mean and variance of X ?

A. $E(X) = \frac{1}{5}, \quad \text{Var}(x) = \frac{4}{25}$

B. $E(X) = \frac{1}{5}, \quad \text{Var}(x) = \frac{4}{5}$

C. $E(X) = \frac{4}{5}, \quad \text{Var}(x) = \frac{4}{25}$

D. $E(X) = \frac{4}{5}, \quad \text{Var}(x) = \frac{4}{5}$

Q4. Fifteen cards are numbered 1 to 19 and placed on a table so that their numbers are hidden. The cards are flipped over one at a time.
What is the minimum number of cards that must be flipped to obtain at least one pair with a difference of 9?

A. 10

B. 11

C. 12

D. 13

Q6. Let $u = x^2 + 4$, the definite integral $\int_0^2 \frac{x}{\sqrt{x^2 + 4}} dx$ can be simplified to

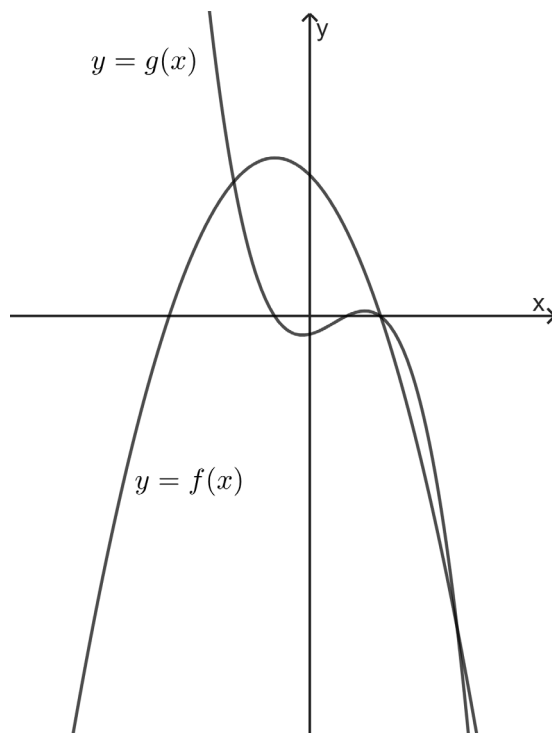
A. $2 \int_4^8 u^{-\frac{1}{2}} du$

B. $\frac{1}{2} \int_4^8 u^{-\frac{1}{2}} du$

C. $2 \int_0^2 u^{-\frac{1}{2}} du$

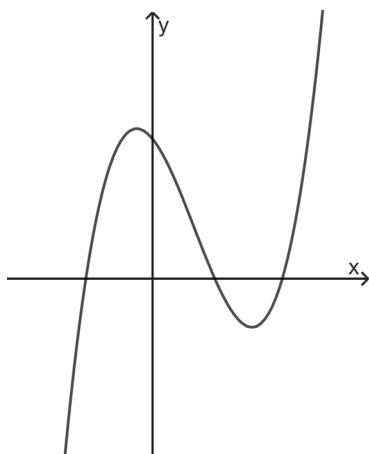
D. $\frac{1}{2} \int_0^2 u^{-\frac{1}{2}} du$

- Q6. The diagram shows the graphs of $y = f(x)$ and $y = g(x)$, where $f(x)$ is a quadratic function and $g(x)$ is a cubic function.

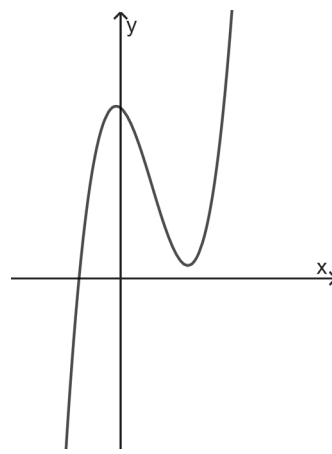


Which of the following best represents the graph of $y = f(x) - g(x)$?

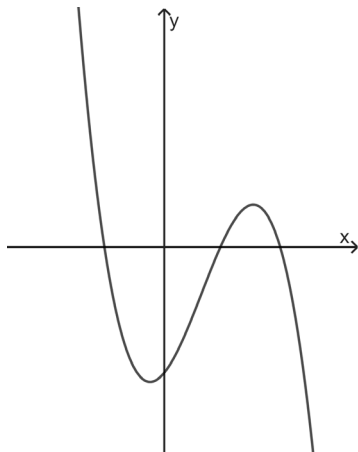
A.



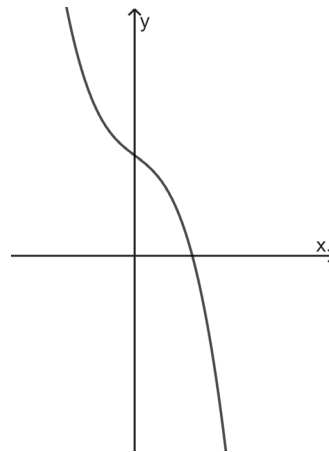
B.



C.



D.



Q7. Which expression is equivalent to $\cos 7x \cos 3x - \sin 8x \sin 4x$?

- A. $\cos 10x - \sin 12x$
- B. $\cos 4x - \sin 4x$
- C. $\sin 11x \sin x$
- D. $\cos 11x \cos x$

Q8. The equation $y = e^{ax}$ satisfies the differential equation $2 \frac{d^2y}{dx^2} + 3 \frac{dy}{dx} - 5y = 0$.

What are the possible values of a ?

- A. $a = 1, a = -\frac{5}{2}$
- B. $a = -1, a = \frac{5}{2}$
- C. $a = 1, a = -\frac{2}{5}$
- D. $a = -1, a = \frac{2}{5}$

Q9. The quartic polynomial $P(x)$ has a zero of 2 and $P(-2) = -4$. Find the remainder when $P(x)$ is divided by $x^2 - 4$.

- A. -4
- B. $x + 2$
- C. $x - 2$
- D. $2 - x$

- Q10. A chocolate factory is designing new products to boost their sales. The graph of $f(x) = 0.8 \cos^{-1}(2x - 1)$, defines a curve that, when rotated about the y -axis will produce a solid that is to be the shape and size of the new chocolate product.

Which integral expression will give the volume of this new chocolate product?

- A. $\frac{\pi}{4} \int_0^{0.8} \left[\cos\left(\frac{5y}{4}\right) + 1 \right]^2 dy$
- B. $\frac{\pi}{4} \int_0^{0.8\pi} \left[\cos\left(\frac{5y}{4}\right) + 1 \right]^2 dy$
- C. $\frac{\pi}{2} \int_0^{0.8} \left[\cos\left(\frac{5y}{4}\right) + 1 \right]^2 dy$
- D. $\frac{\pi}{2} \int_0^{0.8\pi} \left[\cos\left(\frac{5y}{4}\right) + 1 \right]^2 dy$

End of Section I

Section II

60 Marks

Attempt Questions 11–14

Allow about 1 hour and 45 minutes for this section

Answer each question on the relevant pages of the writing booklet. Extra writing booklets are available.

In Questions 11–14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use the Question 11 pages of the Writing Booklet.

- (a) Find the inverse function, $f^{-1}(x)$, of the function $f(x) = 2 - \frac{3}{3-x}$. 2
- (b) Solve $\frac{5-x}{2x-1} \geq 4$. 3
- (c) (i) In how many ways can the letters of the word 'BOOKKEEPER' be arranged? 1
- (ii) What is the probability that a randomly selected arrangement has all the vowels together? 2
- (d) Find the equation of the tangent to the curve $f(x) = \sin^{-1}(2x) - \sqrt{1-4x^2}$ at the point where $x = 0$. 3
- (e) The probability of Jolene scoring a goal in a netball game is 0.65.
- (i) If 10 games are played, what is the probability that she will score exactly 6 goals? Round your answer to 4 significant figures. 2
- (ii) What is the minimum number of games that must be played to ensure the probability of scoring a goal at least once is more than 0.99? 2

End of Questions 11

Question 12 (15 marks) Use the Question 12 pages of the Writing Booklet.

- (a) For the vectors $\mathbf{u} = -7\mathbf{i} - 18\mathbf{j}$ and $\mathbf{v} = \mathbf{i} - 3\mathbf{j}$
- (i) Find the acute angle between $\mathbf{u}$ and $\mathbf{v}$, round to the nearest minute. 2
- (ii) Find a vector that is parallel to $\mathbf{u} - \mathbf{v}$ and has magnitude 3. 2
- (b) Use the substitution $u = 1 - 4x$ to evaluate $\int_{-1}^0 \frac{x dx}{1 - 4x}$. 3
- (c) By expressing $6 \cos \theta + 5 \sin \theta$ in the form $R \cos(\theta - \alpha)$, where $R > 0$, solve the equation $12 \cos \theta + 10 \sin \theta + 1 = 0$ for $0 \leq \theta \leq 2\pi$. Round your answers to 2 decimal places. 4
- (d) The coefficients of the terms in x^3 and x^{-3} in the expansions of $\left(mx + \frac{n}{x^2}\right)^{12}$ are equal, where m and n are non-zero real numbers. Prove that $\frac{m^2}{n^2} = \frac{18}{5}$. 4

End of Questions 12

Question 13 (15 marks) Use the Question 13 pages of the Writing Booklet.

- (a) The polynomial equation $9x^3 - 36x^2 - 52x + 112 = 0$ has roots which form an arithmetic progression. Find all the roots. **3**

- (b) It is given that $\sqrt{16 - x^2} \frac{dy}{dx} = 5y + 10$ and $y = 1$ when $x = 4$. **3**
Find y as a function of x .

- (c) Prove by mathematical induction that **3**
$$\frac{1}{1 \times 7} + \frac{1}{7 \times 13} + \frac{1}{13 \times 19} + \cdots + \frac{1}{(6n - 5)(6n + 1)} = \frac{n}{6n + 1}$$
for integers $n \geq 1$.

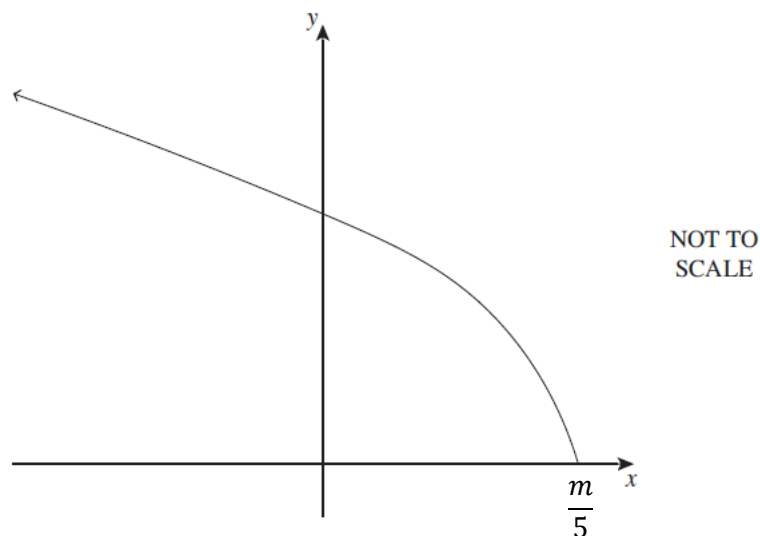
- (d) Let $t = \tan x$, solve the equation $\sin 2x = \tan x$, for $0 \leq x \leq \pi$. **2**

- (e) A box contains ribbons, some of which are red. **4**
150 trials of an experiment are run. In each trial, one ribbon is selected from the bag at random and its colour noted. The ribbon is returned to the box after each trial.
Let X be the random variable representing the number of times a red ribbon is selected.
Given that $E(X) = 60$ and $P(X \geq k) = 0.0062$, find the value of k . You may use the standard normal approximation and the information on page 13.

End of Questions 13

Question 14 (15 marks) Use the Question 14 pages of the Writing Booklet.

- (a) Let $f(x) = \sqrt{m - 5x}$ for $x \leq \frac{m}{5}$. The graph of $f(x)$ is shown. 4



The area enclosed by the graph $f(x)$, the x -axis and y -axis is rotated about the y -axis. Find the value of m such that the volume of the solid formed is $\frac{648\pi}{125} \text{ units}^3$.

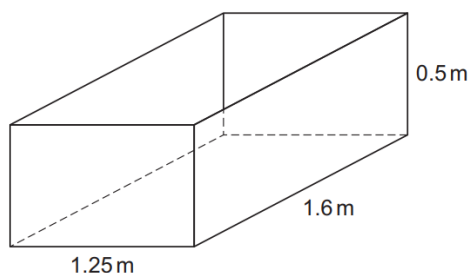
- (b) At 9 am, an ultralight aircraft flies directly over Ayan's head with a perpendicular height of 0.15 km above his eyes. The aircraft maintains a constant speed of 72 km/h and a constant altitude. If x is the horizontal distance travelled by this aircraft and θ is the angle of the elevation from Ayan to the aircraft.

- (i) Show that $\frac{dx}{d\theta} = -\frac{3}{20} \operatorname{cosec}^2 \theta$. 2
- (ii) Hence show that $\frac{d\theta}{dt} = -480 \sin^2 \theta$. 1
- (iii) Find the rate of change of the angle of elevation at 9:01 am. 2

Question 14 continues on next page

Question 14 (continued)

- (c) An empty container, in the shape of a rectangular prism, has length 1.6 metres, width 1.25 metres and depth 0.5 metres, as shown in the diagram below.



The container has a small hole in the bottom. Water is poured into the container at a rate of 0.16 cubic metres per minute.

At time t minutes after the container starts to be filled, the depth of water is d metres and water leak out at a rate of $0.36d^2$ cubic metres per minute.

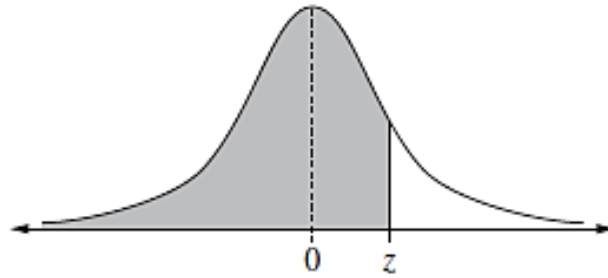
At time t minutes after the container starts to be filled, the volume of water in the container is V cubic metres.

- (i) Show that $\frac{dV}{dt} = \frac{16 - 9V^2}{100}$. 2
- (ii) Given that $\frac{8}{16 - 9x^2} = \frac{1}{4 - 3x} + \frac{1}{4 + 3x}$, find V in terms of t . 3
- (iii) Determine how long it takes to fill the container with water. Give your answer to the nearest minute. 1

End of Paper

Use the information below to answer Question 13 (e)

Table of values $P(Z \leq z)$ for the normal distribution $N(0, 1)$



Z	0.00	0.01	0.02	0.03	0.04	0.05	0.06	0.07	0.08	0.09
0.0	0.5000	0.5040	0.5080	0.5120	0.5160	0.5199	0.5239	0.5279	0.5319	0.5359
0.1	0.5398	0.5438	0.5478	0.5517	0.5557	0.5596	0.5636	0.5675	0.5714	0.5753
0.2	0.5793	0.5832	0.5871	0.5910	0.5948	0.5987	0.6026	0.6064	0.6103	0.6141
0.3	0.6179	0.6217	0.6255	0.6293	0.6331	0.6368	0.6406	0.6443	0.6480	0.6517
0.4	0.6554	0.6591	0.6628	0.6664	0.6700	0.6736	0.6772	0.6808	0.6844	0.6879
0.5	0.6915	0.6950	0.6985	0.7019	0.7054	0.7088	0.7123	0.7157	0.7190	0.7224
0.6	0.7257	0.7291	0.7324	0.7357	0.7389	0.7422	0.7454	0.7486	0.7517	0.7549
0.7	0.7580	0.7611	0.7642	0.7673	0.7704	0.7734	0.7764	0.7794	0.7823	0.7852
0.8	0.7881	0.7910	0.7939	0.7967	0.7995	0.8023	0.8051	0.8078	0.8106	0.8133
0.9	0.8159	0.8186	0.8212	0.8238	0.8264	0.8289	0.8315	0.8340	0.8365	0.8389
1.0	0.8413	0.8438	0.8461	0.8485	0.8508	0.8531	0.8554	0.8577	0.8599	0.8621
1.1	0.8643	0.8665	0.8686	0.8708	0.8729	0.8749	0.8770	0.8790	0.8810	0.8830
1.2	0.8849	0.8869	0.8888	0.8907	0.8925	0.8944	0.8962	0.8980	0.8997	0.9015
1.3	0.9032	0.9049	0.9066	0.9082	0.9099	0.9115	0.9131	0.9147	0.9162	0.9177
1.4	0.9192	0.9207	0.9222	0.9236	0.9251	0.9265	0.9279	0.9292	0.9306	0.9319
1.5	0.9332	0.9345	0.9357	0.9370	0.9382	0.9394	0.9406	0.9418	0.9429	0.9441
1.6	0.9452	0.9463	0.9474	0.9484	0.9495	0.9505	0.9515	0.9525	0.9535	0.9545
1.7	0.9554	0.9564	0.9573	0.9582	0.9591	0.9599	0.9608	0.9616	0.9625	0.9633
1.8	0.9641	0.9649	0.9656	0.9664	0.9671	0.9678	0.9686	0.9693	0.9699	0.9706
1.9	0.9713	0.9719	0.9726	0.9732	0.9738	0.9744	0.9750	0.9756	0.9761	0.9767
2.0	0.9772	0.9778	0.9783	0.9788	0.9793	0.9798	0.9803	0.9808	0.9812	0.9817
2.1	0.9821	0.9826	0.9830	0.9834	0.9838	0.9842	0.9846	0.9850	0.9854	0.9857
2.2	0.9861	0.9864	0.9868	0.9871	0.9875	0.9878	0.9881	0.9884	0.9887	0.9890
2.3	0.9893	0.9896	0.9898	0.9901	0.9904	0.9906	0.9909	0.9911	0.9913	0.9916
2.4	0.9918	0.9920	0.9922	0.9925	0.9927	0.9929	0.9931	0.9932	0.9934	0.9936
2.5	0.9938	0.9940	0.9941	0.9943	0.9945	0.9946	0.9948	0.9949	0.9951	0.9952
2.6	0.9953	0.9955	0.9956	0.9957	0.9959	0.9960	0.9961	0.9962	0.9963	0.9964
2.7	0.9965	0.9966	0.9967	0.9968	0.9969	0.9970	0.9971	0.9972	0.9973	0.9974
2.8	0.9974	0.9975	0.9976	0.9977	0.9977	0.9978	0.9979	0.9979	0.9980	0.9981
2.9	0.9981	0.9982	0.9982	0.9983	0.9984	0.9984	0.9985	0.9985	0.9986	0.9986
3.0	0.9987	0.9987	0.9987	0.9988	0.9988	0.9989	0.9989	0.9989	0.9990	0.9990
3.1	0.9990	0.9991	0.9991	0.9991	0.9992	0.9992	0.9992	0.9992	0.9993	0.9993
3.2	0.9993	0.9993	0.9994	0.9994	0.9994	0.9994	0.9994	0.9995	0.9995	0.9995

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Student Name: _____

Multiple Choice Answer Sheet

Select the alternative A, B, C or D that best answers the question. Fill in the response oval completely.

Sample: $2 + 4 =$ (A) 2 (B) 6 (C) 8 (D) 9
 A ☐ B ☒ C ☐ D ☐

If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

A ☒ B ☒ C ☐ D ☐

If you change your mind and have crossed out what you consider to be the correct answer, then indicate the correct answer by writing the word 'correct' and drawing an arrow as follows.

A ☒ B ☒ C ☐ D ☐
 correct
 ↗

**Start
Here** →

1. A ☐ B ☐ C ☐ D ☐
2. A ☐ B ☐ C ☐ D ☐
3. A ☐ B ☐ C ☐ D ☐
4. A ☐ B ☐ C ☐ D ☐
5. A ☐ B ☐ C ☐ D ☐
6. A ☐ B ☐ C ☐ D ☐
7. A ☐ B ☐ C ☐ D ☐
8. A ☐ B ☐ C ☐ D ☐
9. A ☐ B ☐ C ☐ D ☐
10. A ☐ B ☐ C ☐ D ☐

2026 Mathematics Extension 1 AT4 Trial Solutions

Section 1

Q1	C $f(x) = \tan^{-1}(2x) - \pi$ $-\frac{\pi}{2} < \tan^{-1}(2x) < \frac{\pi}{2}$ $-\frac{3\pi}{2} < \tan^{-1}(2x) - \pi < -\frac{\pi}{2}$ $-\frac{3\pi}{2} < y < -\frac{\pi}{2}$	1 Mark
Q2	A At $x = 0$ $\frac{dy}{dx} = \cos 0 = 1$ This eliminates B and C. At $x = 3$ $\frac{dy}{dx} = \cos 3 \approx -0.99$ Negative gradient matches to A, not D.	1 Mark
Q3	C $P(X = 0) = \frac{3 \times 0 + 1}{5} = \frac{1}{5}$ $P(X = 1) = \frac{3 \times 1 + 1}{5} = \frac{4}{5}$ $E(X) = 0 \times \frac{1}{5} + 1 \times \frac{4}{5} = \frac{4}{5}$ $Var(X) = pq = \frac{1}{5} \times \frac{4}{5} = \frac{4}{25}$	1 Mark
Q4	B The number of pigeonholes (that is, pairs that create a difference of 9): $(1, 10), (2, 11), (3, 12), (4, 13), (5, 14), (6, 15), (7, 16), (8, 17), (9, 18), (10, 19)$ 10 holes By the pigeonhole principle, the eleventh card will guarantee that a pair with a difference of 9 is obtained.	1 Mark
Q5	B $u = x^2 + 4 \rightarrow du = 2x dx$ $x = 2, u = 8$ $x = 0, u = 4$ $\int_0^2 \frac{x}{\sqrt{x^2 + 4}} dx$ $= \frac{1}{2} \int_4^8 \frac{du}{\sqrt{u}}$ $= \frac{1}{2} \int_4^8 u^{-\frac{1}{2}} du$	
Q6	A y-intercept for $f(x)$ is positive and y-intercept for $g(x)$ is negative, therefore $f(x) - g(x)$ is positive and greater than y-intercept for $f(x)$, this eliminates option C. There are one point of intersect to the left of the y-axis and two points of intersection to the right of the y-axis, so $f(x) - g(x) = 0$ for these points, which are the roots of the resulting function. Only option A is correct.	1 Mark

Q7	D $\cos 7x \cos 3x - \sin 8x \sin 4x$ $= \frac{1}{2} [\cos(7x - 3x) + \cos(7x + 3x)] - \frac{1}{2} [\cos(8x - 4x) - \cos(8x + 4x)]$ $= \frac{1}{2} [\cos 4x + \cos 10x] - \frac{1}{2} [\cos 4x - \cos 12x]$ $= \frac{1}{2} (\cos 10x + \cos 12x)$ $= \frac{1}{2} (\cos(11x - x) + \cos(11x + x))$ $= \cos 11x \cos x$	1 Mark
Q8	A $y = e^{ax}$ $\frac{dy}{dx} = ae^{ax}$ $\frac{d^2y}{dx^2} = a^2e^{ax}$ $2\frac{d^2y}{dx^2} + 3\frac{dy}{dx} - 5y = 0$ $2a^2e^{ax} + 3ae^{ax} - 5e^{ax} = 0$ $e^{ax}(2a^2 + 3a - 5) = 0$ $e^{ax}(2a^2 + 5a - 2a - 5) = 0$ $e^{ax}(a(2a + 5) - (2a + 5)) = 0$ $e^{ax}(2a + 5)(a - 1) = 0$ $a = 1, a = -2.5$	1 Mark
Q9	C As 2 is a zero, we have $P(2) = 0$. Dividing $P(x)$ by a quadratic term $(x^2 - 4)$, will leave a remainder at most a linear function. Let this remainder be $ax + b$. $P(x) = (x^2 - 4)Q(x) + ax + b$ $P(2) = 0 = (2^2 - 4)Q(2) + 2a + b$ $2a + b = 0 \dots (1)$ $P(-2) = -4 = ((-2)^2 - 4)Q(2) - 2a + b$ $-2a + b = -4 \dots (2)$ $(1) + (2)$ $2b = -4$ $b = -2$ $2a - 2 = 0$ $a = 1$	1 Mark
Q10	B $y = 0.8 \cos^{-1}(2x - 1)$ $\frac{5y}{4} = \cos^{-1}(2x - 1)$ $2x - 1 = \cos\left(\frac{5y}{4}\right)$ $x = \frac{1}{2} \left(\cos\left(\frac{5y}{4}\right) + 1 \right)$ $x^2 = \frac{1}{4} \left(\cos\left(\frac{5y}{4}\right) + 1 \right)^2$ $0 \leq \cos^{-1}(2x - 1) \leq \pi$ $0 \leq 0.8 \cos^{-1}(2x - 1) \leq 0.8\pi$ $V = \frac{\pi}{4} \int_0^{0.8\pi} \left[\cos\left(\frac{5y}{4}\right) + 1 \right]^2 dy$	1 Mark

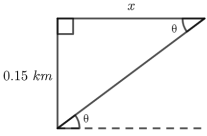
Section 2		
Q11a	$y = 2 - \frac{3}{3-x}$ For $f^{-1}(x)$, swap x and y $x = 2 - \frac{3}{3-y}$ $\frac{3}{3-y} = 2 - x$ $3 = (2-x)(3-y) \quad \text{or} \quad \frac{3}{2-x} = 3 - y$ $3 = 6 - 2y - 3x + xy$ $2y - xy = 3 - 3x \quad y = 3 - \frac{3}{2-x}$ $y(2-x) = 3 - 3x$ $y = \frac{3-3x}{2-x}$ $\therefore f^{-1}(x) = \frac{3-3x}{2-x} = 3 - \frac{3}{2-x} = 3 + \frac{3}{x-2}$	2 Marks Correct solution 1 Mark Swaps x and y and makes significant progress
Q11b	$\frac{5-x}{2x-1} \geq 4, x \neq \frac{1}{2}$ $(5-x)(2x-1) \geq 4(2x-1)^2$ $(5-x)(2x-1) - 4(2x-1)^2 \geq 0$ $(2x-1)(5-x-4(2x-1)) \geq 0$ $(2x-1)(5-x-8x+4) \geq 0$ $(2x-1)(9-9x) \geq 0$ $9(2x-1)(1-x) \geq 0$ $\frac{1}{2} < x \leq 1$	3 Marks Correct solution 2 Marks Makes significant progress 1 Mark Multiplies by the square of the denominator and makes progress
Q11ci	2Os, 2Ks, 3Es $\frac{10!}{2!2!3!} = 151200$	1 Mark Correct solution
Q11cii	Group all the vowels together as one $\frac{6!}{2!} = 360$ Number of ways of arranging the vowels together $\frac{5!}{2! \times 3!} = 10$ $\frac{360 \times 10}{151200} = \frac{1}{42}$	2 Marks Correct solution 1 Mark Finds the number of arrangements of vowels together
Q11d	$f(x) = \sin^{-1}(2x) - \sqrt{1-4x^2}$ $f'(x) = \frac{2}{\sqrt{1-4x^2}} - \frac{1}{2} \times -8x \times (1-4x^2)^{-\frac{1}{2}}$ $f'(x) = \frac{2}{\sqrt{1-4x^2}} + \frac{4x}{\sqrt{1-4x^2}}$ $f'(x) = \frac{2+4x}{\sqrt{1-4x^2}}$ $f'(0) = \frac{2+4 \times 0}{\sqrt{1-4 \times 0}} = 2$ $f(0) = \sin^{-1}(2 \times 0) - \sqrt{1-4 \times 0^2} = -1$ Equation of tangent $y+1 = 2(x-0)$ $2x-y-1 = 0 \text{ or } y = 2x-1$	3 Marks Correct solution 2 Marks Determines the correct gradient 1 Mark Correct differentiation

Q11ei	Probability of scoring is 0.65 Probability of not scoring is $1 - 0.65 = 0.35$ ${}^{10}C_6(0.65)^6(0.35)^4$ $= 0.23766 \dots$ $= 0.2377$ (4 significant figures)	2 Marks Correct solution 1 Mark Finds $(0.65)^6$ or $(0.35)^4$
Q11eii	Scoring at least once is the complement of scoring none. $1 - {}^nC_0(0.65)^0(0.35)^n > 0.99$ $-(0.35)^n > -0.01$ $(0.35)^n < 0.01$ $n \ln 0.35 < \ln 0.01$ $n > \frac{\ln 0.01}{\ln 0.35}$ $n > 4.386 \dots$ $n = 5$ $\therefore$ The least number of games that must be played is 5.	2 Marks Correct solution 1 Mark Establishes $1 - {}^nC_0(0.65)^0(0.35)^n > 0.9$
Q12ai	$\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{ \mathbf{u} \mathbf{v} }$ $\cos \theta = \frac{-7 \times 1 + (-18) \times (-3)}{\sqrt{(-7)^2 + (-18)^2} \times \sqrt{1^2 + (-3)^2}}$ $\cos \theta = \frac{47}{\sqrt{3730}}$ $\theta = 39^\circ 41' 7.64'' = 39^\circ 41'$ (nearest minute)	2 Marks Correct solution 1 Mark Finds $\cos \theta = \frac{47}{\sqrt{3730}}$
Q12aai	$\mathbf{u} - \mathbf{v} = -8\mathbf{i} - 15\mathbf{j}$ $ \mathbf{u} - \mathbf{v} = \sqrt{(-8)^2 + (-15)^2} = 17$ $\frac{3}{17}(-8\mathbf{i} - 15\mathbf{j}) = -\frac{24}{17}\mathbf{i} - \frac{45}{17}\mathbf{j}$	2 Marks Correct solution 1 Mark Finds $ \mathbf{u} - \mathbf{v} $
Q12b	$I = \int_{-1}^0 \frac{x dx}{1 - 4x}$ $u = 1 - 4x \rightarrow x = \frac{1 - u}{4}$ $du = -4dx \rightarrow dx = \frac{1}{-4} du$ $x = 0, u = 1$ $x = -1, u = 5$ $I = \int_5^1 \frac{\frac{1 - u}{4} \times \frac{1}{-4} du}{u}$ $I = \frac{1}{16} \int_1^5 \frac{1 - u}{u} du$ $I = \frac{1}{16} \int_1^5 \left(\frac{1}{u} - 1 \right) du$ $I = \frac{1}{16} [\ln u - u]_1^5$ $I = \frac{1}{16} [(\ln 5 - 5) - (\ln 1 - 1)]$ $I = \frac{1}{16} [\ln 5 - 4] = \frac{\ln 5}{16} - \frac{1}{4}$	3 Marks Correct solution 2 Marks Correct integration 1 Mark Correct u substitution

Q12c	$6 \cos \theta + 5 \sin \theta = R \cos(\theta - \alpha) = R \cos \theta \cos \alpha + R \sin \theta \sin \alpha$ $R \cos \alpha = 6 \dots (1)$ $R \sin \alpha = 5 \dots (2)$ $(1)^2 + (2)^2$ $R^2 \cos^2 \alpha + R^2 \sin^2 \alpha = 6^2 + 5^2$ $R^2 = 61$ $R = \sqrt{61}$ $(2) \div (1)$ $\frac{R \sin \alpha}{R \cos \alpha} = \frac{5}{6}$ $\tan \alpha = \frac{5}{6}$ $\alpha = \tan^{-1} \frac{5}{6} = 0.6947 \dots$ $6 \cos \theta + 5 \sin \theta = \sqrt{61} \cos \left(\theta - \tan^{-1} \frac{5}{6} \right)$ $12 \cos \theta + 10 \sin \theta = -1$ $6 \cos \theta + 5 \sin \theta = -0.5$ $\sqrt{61} \cos \left(\theta - \tan^{-1} \frac{5}{6} \right) = -0.5$ $\cos \left(\theta - \tan^{-1} \frac{5}{6} \right) = -\frac{0.5}{\sqrt{61}}$ $\theta - \tan^{-1} \frac{5}{6} = \pi - \cos^{-1} \left(\frac{0.5}{\sqrt{61}} \right), \pi + \cos^{-1} \left(\frac{0.5}{\sqrt{61}} \right)$ $\theta = \pi - \cos^{-1} \left(\frac{0.5}{\sqrt{61}} \right) + \tan^{-1} \frac{5}{6}, \pi + \cos^{-1} \left(\frac{0.5}{\sqrt{61}} \right) + \tan^{-1} \frac{5}{6}$ $\theta = 2.3295 \dots, 5.3430 \dots$ $\theta = 2.33, 5.34 \text{ (2 d.p.)}$	4 Marks Correct solution 3 Marks Makes significant progress and finds 1 correct θ 2 Marks Correct $R \cos(\theta - \alpha)$ 1 Mark Finds correct R or α
Q12d	$\left(mx + \frac{n}{x^2} \right)^{12} = \binom{12}{0} m^{12} x^{12} + \binom{12}{1} m^{11} x^{11} n^1 x^{-2} + \dots$ $+ \binom{12}{r} m^{12-r} x^{12-r} n^r x^{-2r} + \dots \binom{12}{12} n^{12} x^{-24}$ $\left(mx + \frac{n}{x^2} \right)^{12} = m^{12} x^{12} + 12 m^{11} n^1 x^9 + \dots + \binom{12}{r} m^{12-r} n^r x^{12-3r}$ $+ \dots n^{12} x^{-24}$ For term in x^3, $x^{12-3r} = x^3$ $12 - 3r = 3$ $r = 3$ Coefficient is $\binom{12}{3} m^9 n^3 = 220 m^9 n^3$ For term in x^{-3}, $x^{12-3r} = x^{-3}$ $12 - 3r = -3$ $r = 5$ Coefficient is $\binom{12}{5} m^7 n^5 = 792 m^7 n^5$ $220 m^9 n^3 = 792 m^7 n^5$ $\frac{m^2}{n^2} = \frac{18}{5}$	4 Marks Correct solution 3 Marks Finds correct coefficient for both terms 2 Marks Finds correct coefficient for one term 1 Mark Finds the n^{th} coefficient

Q13a	Let the roots of the polynomial be $\alpha - d, \alpha, \alpha + d$ $(\alpha - d) + \alpha + (\alpha + d) = \frac{36}{9}$ $3\alpha = 4$ $\alpha = \frac{4}{3}$ $(\alpha - d)\alpha(\alpha + d) = \alpha(\alpha^2 - d^2) = -\frac{112}{9}$ $\frac{4}{3}\left(\left(\frac{4}{3}\right)^2 - d^2\right) = -\frac{112}{9}$ $\frac{16}{9} - d^2 = -\frac{28}{3}$ $d^2 = \frac{100}{9}$ $d = \pm \frac{10}{3}$ $\alpha - d = \frac{4}{3} - \frac{10}{3} = -2$ $\alpha + d = \frac{4}{3} + \frac{10}{3} = \frac{14}{3}$ $\therefore \text{The roots are } -2, \frac{4}{3}, \frac{14}{3}$	3 Marks Correct solution 2 Marks Makes significant progress 1 Mark Finds one of the roots
Q12b	$\sqrt{16 - x^2} \frac{dy}{dx} = 5y + 10$ $\int \frac{1}{y + 2} dy = 5 \int \frac{1}{\sqrt{16 - x^2}} dx$ $\ln y + 2 = 5 \sin^{-1} \frac{x}{4} + c$ $y + 2 = Ae^{5 \sin^{-1} \frac{x}{4}}$ $y = Ae^{5 \sin^{-1} \frac{x}{4}} - 2$ $y = 1, x = 4$ $1 = Ae^{5 \sin^{-1} \frac{4}{4}} - 2$ $3 = Ae^{\frac{5\pi}{2}}$ $A = 3e^{-\frac{5\pi}{2}}$ $\therefore y = 3e^{-\frac{5\pi}{2}} \times e^{5 \sin^{-1} \frac{x}{4}} - 2 = 3e^{5 \sin^{-1} \frac{x}{4} - \frac{5\pi}{2}} - 2$	3 Marks Correct solution 2 Marks Correct integration and makes significant progress 1 Mark Correct separation of variables and integrates inverse sine correctly
Q13c	RTP: $\frac{1}{1 \times 7} + \frac{1}{7 \times 13} + \frac{1}{13 \times 19} + \dots + \frac{1}{(6n - 5)(6n + 1)} = \frac{n}{6n + 1}$ 1. Proves true for $n = 1$ $LHS = \frac{1}{1 \times 7} = \frac{1}{7}$ $RHS = \frac{1}{6 \times 1 + 1} = \frac{1}{7}$ $LHS = RHS$ $\therefore \text{True for } n = 1.$ 2. Assume true for $n = k, k$ some positive integer $\frac{1}{1 \times 7} + \frac{1}{7 \times 13} + \frac{1}{13 \times 19} + \dots + \frac{1}{(6k - 5)(6k + 1)} = \frac{k}{6k + 1}$	3 Marks Correct solution 2 Marks Makes significant progress 1 Mark Proves initial case

	3. Prove true for $n = k + 1$ i. e. $\frac{1}{1 \times 7} + \frac{1}{7 \times 13} + \frac{1}{13 \times 19} + \dots + \frac{1}{(6k-5)(6k+1)}$ $+ \frac{1}{(6(k+1)-5)(6(k+1)+1)} = \frac{k+1}{6(k+1)+1}$ $\frac{1}{1 \times 7} + \frac{1}{7 \times 13} + \dots + \frac{1}{(6k-5)(6k+1)} + \frac{1}{(6k+1)(6k+7)} = \frac{k+1}{6k+7}$ $LHS = \frac{1}{1 \times 7} + \frac{1}{7 \times 13} + \dots + \frac{1}{(6k-5)(6k+1)} + \frac{1}{(6k+1)(6k+7)}$ $LHS = \frac{k}{6k+1} + \frac{1}{(6k+1)(6k+7)}$ $LHS = \frac{k(6k+7)+1}{(6k+1)(6k+7)}$ $LHS = \frac{6k^2+7k+1}{(6k+1)(6k+7)}$ $LHS = \frac{(6k+1)(k+1)}{(6k+1)(6k+7)}$ $LHS = \frac{k+1}{6k+7}$ $LHS = RHS$ $\therefore$ Statement is true by mathematical induction for all positive integer n.	
Q13d	$\sin 2x = \tan x, 0 \leq x \leq \pi$ $\frac{2t}{1+t^2} = t$ $2t = t + t^3$ $t^3 - t = 0$ $t(t^2 - 1) = 0$ $t = 0, \pm 1$ $x = 0, \frac{\pi}{4}, \frac{3\pi}{4}, \pi$	2 Marks Correct solution 1 Mark Finds all the t values
Q13e	$E(X) = np$ $150p = 60$ $p = \frac{2}{5}$ $Var(X) = np(1-p)$ $Var(X) = 150 \times \frac{2}{5} \left(1 - \frac{2}{5}\right) = 36$ $P(X \geq k) = P\left(\frac{X-60}{6} \geq \frac{k-60}{6}\right) = P\left(Z \geq \frac{k-60}{6}\right)$ Since $P(X \geq k) = 0.0062$ then $P(X < k) = 1 - 0.0062 = 0.9938$ $P\left(Z \leq \frac{k-60}{6}\right) = 0.9938 = P(Z \leq 2.5)$ $2.5 = \frac{k-60}{6}$ $k = 75$	4 Marks Correct solution 3 Marks Makes significant progress 2 Marks Attempts to use $Z = \frac{X-60}{6}$ as the normal approximation 1 Mark Obtains p value

Q14a	y-intercept, $x = 0$ $y = \sqrt{m - 5 \times 0} = \sqrt{m}$ $y = \sqrt{m - 5x}$ $y^2 = m - 5x$ $5x = m - y^2$ $x = \frac{m - y^2}{5}$ $V = \pi \int_0^{\sqrt{m}} \left(\frac{m - y^2}{5} \right)^2 dy$ $V = \frac{\pi}{25} \int_0^{\sqrt{m}} (m^2 - 2my^2 + y^4) dy$ $V = \frac{\pi}{25} \left[m^2 y - \frac{2my^3}{3} + \frac{y^5}{5} \right]_0^{\sqrt{m}}$ $V = \frac{\pi}{25} \left[\left(m^2 \sqrt{m} - \frac{2m(\sqrt{m})^3}{3} + \frac{(\sqrt{m})^5}{5} \right) - \left(m^2(0) - \frac{2m(0)^3}{3} + \frac{(0)^5}{5} \right) \right]$ $V = \frac{\pi}{25} \left(m^2 \sqrt{m} - \frac{2m^2 \sqrt{m}}{3} + \frac{m^2 \sqrt{m}}{5} \right)$ $V = \frac{\pi}{25} \times \frac{8}{15} m^2 \sqrt{m}$ $V = \frac{8\pi}{375} m^2 \sqrt{m}$ $\frac{8\pi}{375} m^2 \sqrt{m} = \frac{648\pi}{125}$ $m^{\frac{5}{2}} = 243$ $m = 9$	4 Marks Correct solution 3 Marks Correct volume in terms of m 2 Marks Correct volume integration expression 1 Mark Find y-intercept and x in terms of y
Q14bi	 $\tan \theta = \frac{0.15}{x}$ $x = \frac{0.15}{\tan \theta} = 0.15(\tan \theta)^{-1}$ $\frac{dx}{d\theta} = -0.15 \sec^2 \theta (\tan \theta)^{-2}$ $\frac{dx}{d\theta} = -0.15 \times \frac{1}{\cos^2 \theta} \times \frac{\cos^2 \theta}{\sin^2 \theta}$ $\frac{dx}{d\theta} = -\frac{3}{20} \operatorname{cosec}^2 \theta$	2 Marks Correct solution 1 Mark Shows $\tan \theta = \frac{0.15}{x}$
Q14bii	$\frac{d\theta}{dt} = \frac{d\theta}{dx} \times \frac{dx}{dt}$ $\frac{d\theta}{dt} = -\frac{20}{3} \sin^2 \theta \times 72 = -480 \sin^2 \theta$	1 Mark Correct solution

Q14biii	At 9:01 am, $t = \frac{1}{60}$, $x = 72 \times \frac{1}{60} = 1.2 \text{ km}$ Using Pythagoras' theorem, the hypotenuse is $\sqrt{\frac{117}{80}}$ $\sin \theta = \frac{0.15}{\sqrt{\frac{117}{80}}}$ $\frac{d\theta}{dt} = -480 \times \left(\frac{0.15}{\sqrt{\frac{117}{80}}} \right)^2$ $\frac{d\theta}{dt} = -480 \times \frac{1}{65} = -\frac{96}{13} \text{ rad/h}$	2 Marks Correct solution 1 Mark Finds $\sin \theta$
Q14ci	$\frac{dV}{dt} = 0.16 - 0.36d^2$ $V = 1.25 \times 1.6 \times d$ $V = 2d$ $d = \frac{V}{2}$ $\frac{dV}{dt} = 0.16 - 0.36 \times \left(\frac{V}{2} \right)^2$ $\frac{dV}{dt} = \frac{16 - 9V^2}{100}$	2 Marks Correct solution 1 Mark Finds $\frac{dV}{dt} = 0.16 - 0.36d^2$
Q14cii	$\int \frac{8}{16 - 9V^2} dV = \int \frac{2}{25} dt$ $\frac{25}{2} \int \left(\frac{1}{4 - 3V} + \frac{1}{4 + 3V} \right) dV = \int dt$ $\frac{25}{2} \left(-\frac{1}{3} \ln 4 - 3V + \frac{1}{3} \ln 4 + 3V \right) = t + c$ $\frac{25}{6} \left(\ln \left \frac{4 + 3V}{4 - 3V} \right \right) = t + c$ $\frac{4 + 3V}{4 - 3V} = Ae^{\frac{6}{25}t}$ $t = 0, V = 0 \rightarrow A = 1$ $\frac{4 + 3V}{4 - 3V} = e^{\frac{6}{25}t}$ $4 + 3V = 4e^{\frac{6}{25}t} - 3Ve^{\frac{6}{25}t}$ $3V + 3Ve^{\frac{6}{25}t} = 4e^{\frac{6}{25}t} - 4$ $3V \left(1 + e^{\frac{6}{25}t} \right) = 4 \left(e^{\frac{6}{25}t} - 1 \right)$ $\therefore V = \frac{4 \left(e^{\frac{6}{25}t} - 1 \right)}{3 \left(1 + e^{\frac{6}{25}t} \right)}$	3 Marks Correct solution 2 Marks Makes significant progress 1 Mark Correct integration
Q14ciii	$V = 1.25 \times 1.6 \times 0.5 = 1 \text{ m}^3$ $t = \frac{25}{6} \left(\ln \left(\frac{4 + 3 \times 1}{4 - 3 \times 1} \right) \right)$ $t = 8.107 \dots = 8 \text{ minutes}$	1 Mark Correct solution